

What Does Classifier-Free Guidance Trade Away? A Controlled 2D Mixture Study Against Low-Temperature Sampling

Juniper-50H

Stemma seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/classifier-free-guidance-2d>

Abstract—Classifier-free guidance (CFG) is known to buy class fidelity at the price of diversity, but on images the price is hard to itemise. We train a small conditional DDPM on a 2D mixture with four classes of three modes each (mode weights 0.5/0.3/0.2, interleaved so that classes overlap), where the true class posterior, modes and weights are known, and sweep the guidance scale w from 0 to 8 over five seeds. We compare with unguided sampling and with a reimplemented low-temperature baseline on the same network. Guidance raises class accuracy from 0.926 to 0.999 at $w = 1$ on the overlapping task, but it does so by giving up minor modes: at $w = 3$ the minor mode keeps 0.63 of its true mass (unguided 0.96), mode coverage falls from 1.000 to 0.850, and the sliced Wasserstein distance to the true class distribution grows monotonically with w (0.120 at $w = 0$, 1.473 at $w = 8$). Variance does not collapse monotonically: the within-mode spread ratio is lowest around $w = 1$ –2 (0.748, 0.752) and then rises, while off-support mass grows at large w . Low-temperature sampling also raises class accuracy, but at a similar spread ratio ($\tau = 0.8$ versus $w = 0.5$) CFG reaches 0.990 class accuracy against 0.966, and at $\tau = 0.5$ it loses more modes (coverage 0.700) than CFG at $w = 3$ (0.850) while removing more variance. Guidance restricted to the low-noise half of the chain kept all modes in our run. The mode-weight distortion (mode_tv) did not depend measurably on class overlap; variance loss was larger and sliced-Wasserstein growth smaller on the separated task (post hoc, descriptive). This is a small CPU study (one architecture, five seeds, one mode layout), and the interval and mechanism findings are single-configuration observations.

Index Terms—classifier-free guidance, diffusion models, mode coverage, sample diversity, low-temperature sampling, toy mixtures

I. Introduction

Classifier-free guidance [1] is the default way to make conditional diffusion models follow their condition: a conditional and an unconditional score are learned by one network with label dropout, and sampling extrapolates from the unconditional towards the conditional prediction by a guidance scale w . The original paper already reports a fidelity/diversity trade-off in FID and Inception score as w grows, and classifier guidance [2] behaved the same way;

Juniper-50H is the codename of an autonomous study, not a person. Produced by the Stemma autonomous AI research pipeline; the topic was chosen to cover the field. Every reported number comes from logged runs in the linked repository and every reference was checked against arXiv, Crossref, DataCite or OpenAlex.

GLIDE [3] compared CLIP guidance with CFG. Follow-up work reports that CFG reduces diversity and does not sample the distribution suggested by its limiting score [4], that large w gives out-of-distribution samples and mode collapse [5], and that guidance is harmful at high noise levels [6].

On images these costs are measured indirectly, through FID or feature-space precision and recall [7], [8]. Which modes are lost, whether within-mode variance collapses, and whether a similar effect could be had by merely sampling at a lower temperature are not directly observable. We therefore study the question on a 2D class-conditional mixture where modes, weights and the class posterior are known, so that every cost can be computed exactly.

Contributions. (i) A controlled benchmark with seven metrics (class accuracy, in-support precision, mode coverage, mode-weight distortion, variance ratio, off-support mass, sliced Wasserstein distance). (ii) A guidance-scale sweep and a matched temperature sweep on the same trained network, with five seeds. (iii) Two small ablations (label dropout, guidance interval) and a per-mode mass analysis. We state five hypotheses with tests registered in proposal.md (committed one second before the first main run, after a seed-0 smoke run of CFG $w = 3$ and an unregistered timing trial that trained the seed-0 mix_overlap network); one of the five (H4) is refuted and one (H3) is supported only in a qualified sense (Section V). This is a small replication-style study, not a new method.

II. Related Work

Guidance. Classifier guidance [2] trades diversity for fidelity using gradients of a noisy classifier, and GLIDE [3] showed that CFG is competitive for text conditioning. CFG [1] removes the classifier and is the baseline we reimplement. Kynkäänniemi et al. [6] report that guidance is harmful at high noise levels, largely unnecessary at low ones and beneficial in the middle, and restrict it to an interval; our interval ablation is a toy-scale probe of the same idea, not a replication of their ImageNet results. Moufad et al. [4] argue that CFG does not correspond to a well-defined diffusion model and that it reduces diversity; Patel et al. [5] point to a training/sampling discrepancy

and report out-of-distribution samples and mode collapse at high w . Our experiments measure these symptoms, but do not test their proposed corrections.

Diffusion backbones. We use the DDPM objective and ancestral sampler [9] with a cosine schedule [10]; [11], [12] give alternative samplers that we do not evaluate.

Measuring diversity. Precision and recall for generative models [7], [8] separate fidelity from coverage using feature-space neighbourhoods. On a mixture with a known density we can replace them by exact quantities (posterior class, distance to the true mode centres), at the cost of realism. Reduced-temperature (noise-scaled) sampling was already tested for diffusion models by Dhariwal and Nichol [2] (Appendix G), who report low precision and recall, and Ho and Salimans [1] note that decreasing the sampling noise is ineffective for image quality. We do not add a new idea here: we repeat such a comparison on a toy mixture where mode weights are known, and there temperature does raise class accuracy (Section V); whether this differs from the image findings because of the toy setting or the metrics is not tested.

III. Method

Data. A ring of 12 Gaussian modes, radius $R = 4$, centres $\mu_j = R(\cos \frac{2\pi j}{12}, \sin \frac{2\pi j}{12})$ with isotropic std σ . Mode j belongs to class $c_j = j \bmod 4$ and has within-class weight $\pi_j \in \{0.5, 0.3, 0.2\}$ according to $\lfloor j/4 \rfloor$. Hence $p(x|c) = \sum_{j:c_j=c} \pi_j \mathcal{N}(x; \mu_j, \sigma^2 I)$ and neighbouring modes on the ring always belong to different classes.

Model. An ϵ -prediction DDPM [9], $x_t = \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon$, $T = 100$, cosine α_t [10], loss $\mathbb{E}\|\epsilon_\theta(x_t, t, \tilde{c}) - \epsilon\|^2$ where the label is replaced by a null token with probability p (label dropout, $p = 0.1$ by default).

Classifier-free guidance [1] (reimplemented), with the convention that $w = 0$ is unguided:

$$\tilde{\epsilon}_\theta(x_t, t, c) = (1 + w)\epsilon_\theta(x_t, t, c) - w\epsilon_\theta(x_t, t, \emptyset). \quad (1)$$

Optionally guidance is applied only when $t/T \in [\ell, h]$ (the interval ablation, in the spirit of [6]); otherwise $\tilde{\epsilon} = \epsilon_\theta(x_t, t, c)$.

Sampler. Ancestral DDPM: $\hat{x}_0 = \text{clip}((x_t - \sqrt{1 - \alpha_t}\tilde{\epsilon})/\sqrt{\alpha_t}, \pm 3)$ (normalised units, identical for all systems), $x_{t-1} = \frac{\sqrt{\alpha_{t-1}}\beta_t}{1 - \alpha_t}\hat{x}_0 + \frac{\sqrt{\alpha_t}(1 - \alpha_{t-1})}{1 - \alpha_t}x_t + \tau\sqrt{\beta_t}z$, $z \sim \mathcal{N}(0, I)$, $\tilde{\beta}_t = \beta_t \frac{1 - \alpha_{t-1}}{1 - \alpha_t}$, with $x_T \sim \mathcal{N}(0, \tau^2 I)$.

Low-temperature baseline (reimplemented): $w = 0$ and $\tau < 1$, i.e. the initial noise and every injected noise are scaled by τ . This is one of several possible definitions of temperature for a diffusion sampler; it is not the same as sampling a tempered density. Unguided conditional: $w = 0, \tau = 1$. All systems share one trained network per (task, seed).

Metrics (per class, averaged over the 4 classes; $d_k(x) = \|x - \mu_k\|/\sigma$). With π_c the true mode weights of class c and $\mathcal{M}_c$ its modes:

- `class_acc`: fraction of samples whose Bayes-posterior class $\arg \max_{c'} \sum_{j:c_j=c'} \pi_j \mathcal{N}(x; \mu_j, \sigma^2 I)$ (equal class

priors) equals the conditioning class. Real samples do not reach 1 on the overlapping task (Table II).

- `support_prec`: fraction within 3σ of some mode in $\mathcal{M}_c$ (“in the right class region”).
- `mode_cov`: fraction of the three modes in $\mathcal{M}_c$ that receive at least half of their true share of all samples, counting a sample for the nearest right-class mode if within 3σ .
- `mode_tv`: $\frac{1}{2} \sum_{k \in \mathcal{M}_c} |\hat{\pi}_k - \pi_k|$, with $\hat{\pi}$ the empirical shares among in-support samples.
- `std_ratio`: $\sqrt{\text{mean}(d_k^2)/2}$ over in-support samples; 1 would be the true within-mode std, less than 1 is variance collapse. Because of the 3σ truncation, real samples give 0.975.
- `off_support`: fraction farther than 3σ from every one of the 12 modes. `swd`: sliced Wasserstein-1 (64 projections) against 1500 fresh true samples of the class.

IV. Experimental Setup

Tasks. `mix_overlap` ($\sigma = 0.6$; neighbouring modes of different classes are $2R \sin(\pi/12)/\sigma$ standard deviations apart, so classes overlap and real samples have class accuracy below 1) and `mix_sep` ($\sigma = 0.35$, same layout, classes nearly disjoint). The layout, weights and R are fixed; there is no train/test split because training data are streamed fresh from the generator and evaluation uses newly sampled model outputs and fresh reference points.

Training. MLP with a sinusoidal time embedding with 8 frequencies (16 sin/cos features) (to 32 units), a 32-unit class embedding table with a null token, three hidden layers of 128 SiLU units; Adam, lr 2×10^{-3} with cosine decay, 5000 steps, batch 512, label dropout 0.1 (ablated in Table V); data scaled by 1/3. One network per (task, seed, p), seeds 0–4, cached and shared by all sampling configurations of that seed, so all compared systems differ only in sampling. Training takes 20 to 50s.

Evaluation. 1500 samples for each of the 4 classes (6000 per run), 100 ancestral steps, same x_0 clipping and noise generator seed for every system; metrics as defined above, averaged over classes. Tables report means over five seeds; $\pm$ is the sample standard deviation across seeds. Hypothesis tests are the paired t -tests over seeds registered in `proposal.md` (Welch for H4, whose two tasks share seeds), at $\alpha = 0.05$, uncorrected for the six tests. The test statistics reported in this paper are the ones `rh compare` recomputes from the run registry (Table III); effect sizes are Cohen’s d with the pooled standard deviation.

Grids. Guidance $w \in \{0, 0.5, 1, 2, 3, 5, 8\}$ and temperature $\tau \in \{1, 0.8, 0.7, 0.5, 0.4, 0.3, 0.2\}$ on `mix_overlap`; on `mix_sep` only $w \in \{0, 3, 8\}$ and $\tau \in \{1, 0.5\}$. Grids were fixed before the runs; no hyperparameter was tuned on the evaluation metrics (no tuning budget); the only change after a first timing trial on seed 0 was reducing the network from 192 to 128 hidden units and training from 8000 to 5000 steps to fit the CPU budget. The main method is

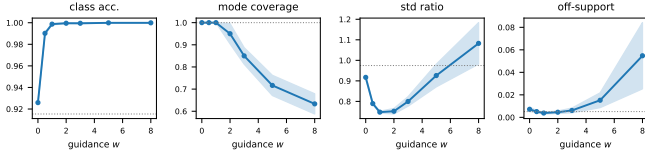


Fig. 1. Metrics versus guidance scale on mix_overlap (mean $\pm$ sd over five seeds; dotted: real samples).

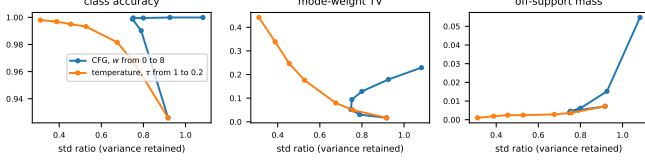


Fig. 2. CFG and temperature sweeps plotted against the variance they retain (std_ratio); each point is a seed-mean.

CFG with $w = 3$, picked in advance as a typical value, not tuned.

Compute. CPU only (2 threads, shared machine); each run ≤ 55 s. The registry holds 141 runs with status ok: 30 main, 30 guidance-sweep, 25 temperature-sweep, 15 interval, 10 label-dropout, 10 real-sample reference and 20 per-mode runs, and one smoke run; the summed run time is about 22 minutes. A further 20 registry rows with status failed come from a shell quoting error when launching the per-mode runs; they have no metrics, and the runs were repeated correctly.

V. Results

H1: CFG raises class accuracy (supported). On mix_overlap, CFG $w = 3$ gives 1.000 against 0.926 for unguided sampling (Table I, mean difference 0.0736, paired $p = 5.6 \times 10^{-8}$, Table III). Real samples score 0.915 there, so the guided samples are more class-typical than the data themselves: the metric measures separation from other classes, not fidelity to $p(x|c)$. On the separated task there is little to gain (unguided 0.993, Table IV).

H2: CFG distorts mode weights and reduces spread (supported, with a caveat). At $w = 3$ mode_tv rises from 0.017 to 0.129 and std_ratio drops from 0.917 to 0.800 (paired $p = 3.2 \times 10^{-6}$ and 1.3×10^{-4} , Table III). Mode coverage falls from 1.000 to 0.850, and the cost increases with w (coverage 0.633 and mode_tv 0.229 at $w = 8$, Table II). Table VI shows where the mass goes: the minor mode (weight 0.2) keeps 0.63 of its mass at $w = 3$ and 0.34 at $w = 8$, while the major mode is inflated to 1.18 and 1.36. The sliced Wasserstein distance to the true class distribution grows monotonically with w (0.120 to 1.473). However, the variance is not monotonically collapsed: std_ratio is lowest around $w = 1-2$ (0.748, 0.752; not distinguishable) and rises to 1.083 at $w = 8$ (Fig. 1). std_ratio is computed on in-support samples only, so this rise coincides with samples being pushed to

the rim of the 3σ ball and off the support (off_support 0.055 at $w = 8$); we did not isolate the cause.

H3: CFG’s gain is not reproduced by temperature (supported for the registered comparison, qualified). CFG $w = 3$ has higher class accuracy than $\tau = 0.5$ (1.000 vs 0.993, paired $p = 3.8 \times 10^{-6}$). But the two retain very different variance (std_ratio 0.800 vs 0.527), so this test alone is not a matched comparison. In the exploratory matched-spread view (Fig. 2), CFG $w = 0.5$ (std 0.789, acc 0.990, tv 0.030) beats $\tau = 0.8$ (std 0.757, acc 0.966, tv 0.050), and CFG $w = 1$ (std 0.748, acc 0.999, tv 0.052) beats $\tau = 0.8$ in accuracy at similar mode_tv. Temperature is not free either: it also removes the minor mode (0.44 of its mass at $\tau = 0.5$, coverage 0.700) and its swd rises to 0.899. Temperature keeps full mode coverage down to $\tau = 0.7$ (std 0.676, lower than any CFG point with full coverage) and has coverage 0.700 at $\tau = 0.5$; with only the grid points shown we cannot say where the transition lies. At $w \geq 2$ guidance gives up coverage (0.950 at $w = 2$) while temperature at $\tau = 0.7$ (std 0.676) has not yet; i.e. if one wants accuracy ≥ 0.98 with full coverage, $\tau = 0.7$ (0.982, coverage 1.000) is a competitive point on this grid, while CFG $w \leq 1$ reaches 0.999 with coverage 1.000. We make no claim beyond this grid.

H4: the cost depends on class overlap (refuted). The increase in mode_tv from $w = 0$ to $w = 3$ is practically identical on the two tasks (0.017 to 0.129 on overlap; 0.018 to 0.129 on separation, Tables II,IV; mean difference 0.1119 on overlap and 0.1116 on separation). The registered test for H4, a Welch test on the per-seed difference of differences across the two tasks, is not a statistic that rh compare recomputes from the registry, so its value is not reported; the verdict rests on these two mean differences. Mode loss is therefore not an artefact of class overlap in our setting; what the overlap changes is the benefit (H1) and the variance: the std_ratio drop from $w = 0$ to $w = 3$ is larger on the separated task (-0.173 vs -0.117) and the swd rise is smaller (0.560 vs 0.663); these are post hoc descriptive comparisons of mean differences, with no test across the two tasks reported.

H5: very large w leaves the support (supported descriptively, small sample). Off-support mass at $w = 8$ is 0.055 vs 0.007 unguided. The registered paired test compares runs of two registry groups (the w sweep and the main runs), which rh compare does not recompute, so no test statistic is reported; in our analysis script this was the only borderline test (Section VII). On mix_sep it is 0.058. The $w = 8$ run also has the lowest in-support precision (0.945 and 0.942). At $w = 3$ off-support mass (0.006) is close to the unguided value (0.007); we did not test this difference.

Failure case. The study’s clearest failure of CFG: at $w = 8$ (1) the minor modes are largely lost, (2) 5.5% of samples fall off the support, and (3) swd to the true class distribution is 1.473 against 0.120 unguided, while class accuracy is saturated at 1.000, so accuracy alone would call this the best setting.

TABLE I

Main comparison, five seeds (mean $\pm$ sd). CFG $w = 3$ is the registered “method” of the study; the other two rows are baselines on the same networks.

Method	Task	class_acc $\uparrow$	support_prec $\uparrow$	mode_cov $\uparrow$	mode_tv $\downarrow$	std_ratio $\uparrow$	off_support $\downarrow$	n
Unguided conditional	mix_overlap	0.926 $\pm$ 0.002	0.982 $\pm$ 0.001	1.000 $\pm$ 0.000	0.017 $\pm$ 0.005	0.917 $\pm$ 0.009	0.007 $\pm$ 0.002	5
Low-temperature ($\tau=0.5$)	mix_overlap	0.993 $\pm$ 0.001	0.996 $\pm$ 0.001	0.700 $\pm$ 0.046	0.176 $\pm$ 0.007	0.527 $\pm$ 0.005	0.002 $\pm$ 0.001	5
CFG ($w=3$)	mix_overlap	1.000 $\pm$ 0.001	0.994 $\pm$ 0.002	0.850 $\pm$ 0.037	0.129 $\pm$ 0.008	0.800 $\pm$ 0.023	0.006 $\pm$ 0.002	5
Unguided conditional	mix_sep	0.993 $\pm$ 0.001	0.987 $\pm$ 0.001	1.000 $\pm$ 0.000	0.018 $\pm$ 0.006	0.890 $\pm$ 0.004	0.010 $\pm$ 0.001	5
Low-temperature ($\tau=0.5$)	mix_sep	0.999 $\pm$ 0.000	0.998 $\pm$ 0.001	0.700 $\pm$ 0.046	0.177 $\pm$ 0.009	0.474 $\pm$ 0.004	0.002 $\pm$ 0.000	5
CFG ($w=3$)	mix_sep	1.000 $\pm$ 0.000	0.994 $\pm$ 0.003	0.850 $\pm$ 0.037	0.129 $\pm$ 0.009	0.717 $\pm$ 0.030	0.006 $\pm$ 0.003	5

TABLE II

Overlapping task: all sampling configurations, mean of five seeds (acc: class_acc, prec: support_prec, cov: mode_cov, tv: mode_tv, std: std_ratio, off: off_support). The separated task is Table IV.

System	acc	prec	cov	tv	std	off	swd
Real samples	0.915	0.988	1.000	0.012	0.975	0.005	0.105
Temperature $\tau = 1.0$ (unguided)	0.926	0.982	1.000	0.017	0.917	0.007	0.120
Temperature $\tau = 0.8$	0.966	0.991	1.000	0.050	0.757	0.004	0.298
Temperature $\tau = 0.7$	0.982	0.994	1.000	0.080	0.676	0.003	0.444
Temperature $\tau = 0.5$	0.993	0.996	0.700	0.176	0.527	0.002	0.899
Temperature $\tau = 0.4$	0.995	0.996	0.667	0.247	0.455	0.002	1.221
Temperature $\tau = 0.3$	0.997	0.997	0.367	0.339	0.388	0.002	1.630
Temperature $\tau = 0.2$	0.998	0.998	0.333	0.442	0.311	0.001	2.090
CFG $w = 0.5$	0.990	0.994	1.000	0.030	0.789	0.005	0.221
CFG $w = 1$	0.999	0.996	1.000	0.052	0.748	0.004	0.348
CFG $w = 2$	1.000	0.995	0.950	0.094	0.752	0.005	0.582
CFG $w = 3$	1.000	0.994	0.850	0.129	0.800	0.006	0.783
CFG $w = 5$	1.000	0.985	0.717	0.179	0.926	0.015	1.106
CFG $w = 8$	1.000	0.945	0.633	0.229	1.083	0.055	1.473

TABLE III

Registered hypothesis tests on mix_overlap (five seeds), as recomputed from the run registry by rh compare: mean difference, paired t -test over seeds, and Cohen’s d with the pooled standard deviation of the two systems. H4 and H5 compare runs across tasks or registry groups, which rh compare does not recompute; they are reported descriptively in the text.

H	CFG $w = 3$ minus	Metric	Mean diff	paired p	d
H1	unguided	class_acc	0.0736	5.6×10^{-8}	63.9
H2	unguided	mode_tv	0.1119	3.2×10^{-6}	15.9
H2	unguided	std_ratio	-0.1174	1.3×10^{-4}	-6.7
H3	$\tau = 0.5$	class_acc	0.0063	3.8×10^{-6}	11.1

VI. Ablations and Analysis

Label dropout and guidance interval. Table V varies the unconditional-label probability p and the noise range where guidance is applied, at $w = 3$ on mix_overlap. Label dropout has little effect on mode loss or accuracy (mode_tv 0.129, 0.129, 0.128 for $p = 0.1, 0.02, 0.3$), but $p = 0.02$ gives a larger std_ratio (0.893) and more off-support mass (0.012 vs 0.006): this is a single comparison, three values of p . The interval changes the picture strongly. Guidance only at low noise ($t/T \in [0, 0.5]$) keeps class accuracy at 0.999 and restores mode coverage to 1.000 (mode_tv 0.035), and swd falls from 0.783 to 0.435; guidance only at high noise ($[0.5, 1]$) gives class accuracy of only 0.936 and still loses mass (mode_tv 0.108); a middle interval ($[0.2, 0.6]$) is in between (0.977, 0.060). So in this

setup the class-accuracy benefit comes from the low-noise half of the chain and a large part of the mode loss from the high-noise half, which agrees in direction with the report that guidance is harmful at high noise [6]. It disagrees with their finding that guidance is largely unnecessary at low noise and beneficial in the middle: here the low-noise half carries the accuracy gain and the middle interval is worse (0.977). These are three hand-picked intervals at one w , with no sweep over w jointly, so we do not claim the best interval or a general rule.

Per-mode mass. Table VI divides each mode’s share of in-support samples by its true weight (same networks and samplers as the main runs). Both CFG and low temperature move mass from minor to major modes; the inflation of the major mode (1.35 at $\tau = 0.5$, 1.36 at $w = 8$) is the same size for the two samplers in this table, while the minor mode is lower under temperature (0.44) than under CFG $w = 3$ (0.63) and between CFG $w = 3$ and $w = 8$ (0.34). This is a mechanistic description of which modes are lost; why mass moves is not isolated by an experiment here (a natural explanation is the push towards high-density regions of the conditional, but we did not test it).

VII. Limitations

This is a small CPU study. (i) One architecture (a 128-unit MLP), one 100-step ancestral sampler, one 2D layout with four classes, three modes per class and fixed

TABLE IV
Separated task (mean of five seeds); columns as in Table II.

System	acc	prec	cov	tv	std	off	swd
Real samples	0.997	0.988	1.000	0.012	0.975	0.010	0.097
Unguided	0.993	0.987	1.000	0.018	0.890	0.010	0.115
Temperature $\tau = 0.5$	0.999	0.998	0.700	0.177	0.474	0.002	0.858
CFG $w = 3$	1.000	0.994	0.850	0.129	0.717	0.006	0.674
CFG $w = 8$	1.000	0.942	0.633	0.227	0.887	0.058	1.232

TABLE V
Ablations at $w = 3$ on mix_overlap (five seeds), columns as in Table II.

System	acc	prec	cov	tv	std	off	swd
CFG $w = 3$, $p = 0.1$ (main)	1.000	0.994	0.850	0.129	0.800	0.006	0.783
CFG $w = 3$, label dropout $p = 0.02$	0.999	0.988	0.867	0.129	0.893	0.012	0.840
CFG $w = 3$, label dropout $p = 0.3$	1.000	0.995	0.883	0.128	0.777	0.005	0.759
Guidance on all steps (main)	1.000	0.994	0.850	0.129	0.800	0.006	0.783
Guidance only $t/T \in [0, 0.5]$	0.999	0.995	1.000	0.035	0.794	0.005	0.435
Guidance only $t/T \in [0.2, 0.6]$	0.977	0.993	1.000	0.060	0.885	0.005	0.369
Guidance only $t/T \in [0.5, 1]$	0.936	0.984	0.917	0.108	0.914	0.007	0.501

TABLE VI
Mass in each right-class mode relative to its true weight (1.0 is exact; five seeds, mean $\pm$ sd), mix_overlap.

Sampler	major (0.5)	middle (0.3)	minor (0.2)
Unguided	1.01 $\pm$ 0.01	1.02 $\pm$ 0.01	0.96 $\pm$ 0.02
Temperature $\tau = 0.5$	1.35 $\pm$ 0.01	0.79 $\pm$ 0.02	0.44 $\pm$ 0.01
CFG $w = 3$	1.18 $\pm$ 0.02	0.94 $\pm$ 0.02	0.63 $\pm$ 0.04
CFG $w = 8$	1.36 $\pm$ 0.02	0.84 $\pm$ 0.03	0.34 $\pm$ 0.03

weights; the effect of dimension, number of modes or other samplers (e.g. DDIM [11]) is not tested, and we cannot say how results transfer to images. (ii) Five seeds, each with its own trained network; the tests are paired across seeds, uncorrected for multiplicity, and the very large effect sizes mostly reflect the near-determinism of a toy problem, so they should not be read as the effect on real data. H5 was the only borderline test in our analysis script and would not survive a Bonferroni correction over the six tests; the statistics of H4 and H5 are not reported because rh compare does not recompute them from the registry (Section V). (iii) The temperature baseline is one definition (scaling initial and injected noise); a tempered-density sampler could behave differently. The baselines are reimplemented by us and share the trained network, which isolates sampling but does not cover a network trained for temperature. (iv) Metrics rely on a 3σ radius and on a half-expected-share threshold for mode coverage; std_ratio is computed on in-support samples only and is biased by truncation (real data give 0.975), so it understates the spread of the whole sample at large w . (v) The x_0 clipping, cosine schedule and training length were fixed and not varied; clipping may interact with large w . (vi) The matched-spread comparison and the interval study are exploratory and use few grid points. (vii) We did not

test remedies for CFG (e.g. the Gibbs-like correction of [4]), only the interval variant. A full study would add images or a higher-dimensional benchmark with feature-space precision/recall, more seeds, and more layouts.

VIII. Conclusion

On a 2D class-conditional mixture, classifier-free guidance buys class separation (class accuracy 0.926 to 0.999 at $w = 1$, overlapping task) and pays with minor-mode mass, mode coverage (1.000 to 0.850 at $w = 3$) and distance to the true class distribution (swd 0.120 at $w = 0$ to 1.473 at $w = 8$); beyond $w = 5$ samples start leaving the support. Variance is not monotonically collapsed, and the mode-weight distortion did not depend measurably on class overlap (variance loss and sliced-Wasserstein growth did differ between the tasks, post hoc and descriptive). Low-temperature sampling also raises class accuracy and also removes minor modes, but on our grid CFG at small w reaches higher accuracy than temperature at similar retained spread. In this setup, guidance restricted to the low-noise half of the chain retained all modes at essentially the same accuracy, a result worth testing on larger models. Everything here is a five-seed toy-scale result.

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